M13D — Mock exam 2

Instruments · Rotary Wing Aerodynamics · Autoflight · Equipment and Furnishings

40 questions · 20 Instruments · 13 Autoflight · 5 Rotary Wing Aerodynamics · 2 Equipment and Furnishings

Level 3: 22 · Level 2: 13 · Level 1: 5. Three options, one correct.

Every question sits on a fact highlighted or underlined in the manual. No question in this paper appears in any of the other four.

The answer key follows the questions, with the explanation and the manual PDF page for each one.

INSTRUMENTS

1. An instrument is to be fitted in the middle at the top of a Basic T panel and must also display the indication from the flight director system. Which indicator is required?

- **A.** The HSI
- **B.** The ADI
- **C.** The turn coordinator

2. An aircraft can no longer hold its selected flight level unless the pilot flies manually. Which requirement for flight in RVSM airspace is no longer met?

- **A.** An autopilot that can maintain the altitude
- **B.** A device for indicating when leaving the predetermined height
- **C.** A transponder with altitude transmission

3. In a Bourdon tube gauge the movement of the tube tip is proportional to

- **A.** The applied pressure
- **B.** The tube temperature
- **C.** The signal frequency

4. Two lines are parted at a quick-disconnect during maintenance. What happens at the coupling?

- **A.** It stays open until a cap is fitted
- **B.** It vents the line to ambient air
- **C.** It seals itself automatically

5. The same static line leak is considered in an unpressurized aircraft and in a pressurized one. How does the altimeter behave in each case?

- **A.** It decreases the indication in the unpressurized aircraft and slightly increases it in the pressurized one
- **B.** It slightly increases the indication in the unpressurized aircraft and decreases it in the pressurized one
- **C.** It slightly increases the indication in both aircraft

6. The pitot tube becomes totally blocked in cruise and the thrust is then increased to accelerate. What does the airspeed indicator show?

- **A.** No change in the indicated speed
- **B.** A rise in the indicated speed
- **C.** A fall in the indicated speed to zero

7. The temperature indication is needed in order to determine

- **A.** IAS
- **B.** EAS
- **C.** TAS

8. In flight, how do the three measured air temperature values compare with one another?
   - A. SAT is the highest, RAT is lower than SAT but higher than TAT
   - B. TAT is the highest, RAT is lower than TAT but higher than SAT
   - C. RAT is the highest, TAT is lower than RAT but higher than SAT

9. A hybrid air data computer is described as hybrid because
   - A. it uses both pressure capsules and pressure transducers
   - B. it can be supplied with either AC or DC power
   - C. it provides outputs in analogue as well as digital form

10. An air data module at a pitot probe has converted the pressure it senses. Which path does the resulting signal then follow?
   - A. To an air data inertial reference unit, and from there on to the user systems
   - B. To the display management computers, and from there on to an air data inertial reference unit
   - C. To the standby instruments, and from there on to an air data inertial reference unit

11. An angle of attack sensor is a tube lying crosswise to the air stream, pierced by two rows of holes. How does it sense the angle?
   - A. by measuring the total pressure at the forward row
   - B. by sensing the movement of the stagnation point between the rows
   - C. by comparing the pressures sensed at the two rows

12. Besides recording and storing digital flight data, the wireless quick access recorder is also able to
   - A. transmit the digital flight data to the ground
   - B. format the digital flight data for the recorder
   - C. erase the digital flight data after transfer

13. A main gearbox has no cockpit instrument or warning of its own. Why does that make its speed probe reading so valuable?
   - A. Because a vibration there could otherwise be felt through the controls
   - B. Because an overheat there could otherwise be read on the oil gauge
   - C. Because an overspeed there could otherwise go unidentified by the crew

14. A commercial aircraft is fitted with electrical directional gyros. How is their drift dealt with?
   - A. It is compensated by the pilot each hour
   - B. It is compensated automatically
   - C. It is compensated by the vertical gyro

15. Some error is still left after a compass has been compensated — where is that remainder written down?
   - A. on a label fitted beside the flux valve
   - B. on the compensating disc at the airport site
   - C. on a deviation card carried on the aircraft

16. A pushbutton is lit in a colour that reports normal operation. Which two colours can be used for that?
   - A. Red or amber
   - B. Amber or white
   - C. Green or blue

17. How many chimes are sounded for a level 2 caution
   - A. None
   - B. Three
   - C. One

18. A recorder fault and a stall warning occur on the same flight. How do the two alerts differ in what the crew receives?

- **A.** One gives a single chime while the other gives no sound
- **B.** Each one gives an aural alarm but only one gives a light
- **C.** One gives no sound or light while the other gives both

19. The limits of one of the alerting envelopes are exceeded. What does the enhanced ground proximity warning system give the flight crew?

- **A.** Aural alert messages with visual annunciations and displays
- **B.** An automatic pull-up commanded through the autopilot
- **C.** A master warning that has to be reset by the crew

20. A pilot reads the rotor rpm on the triple tachometer. Which part of the dial shows it?

- **A.** The large pointer N1 on the outer scale
- **B.** The digital window at the foot of the dial
- **C.** The small pointer R on the inner scale

AUTOFLIGHT

21. Which motions are described by flight condition parameters

- **A.** oscillatory motions
- **B.** rotary motions
- **C.** translatory motions

22. Vertical speed is treated as a basic mode by some autopilots. Why is this called an exception?

- **A.** Vertical speed is in fact a flight path parameter
- **B.** Vertical speed is in fact named after a flight phase
- **C.** Vertical speed is in fact a flight condition parameter

23. Whether an overpowering of the flight controls leads to disengagement depends on

- **A.** The mode selected and the channel that is engaged
- **B.** The circumstances of the input and the autopilot itself
- **C.** The torque the limiter is built to pass and the airspeed

24. During the synchronization phase the pitch regulator produces no actuator activity at all. Why is that so?

- **A.** The regulator output is clamped, so the actuator cannot respond
- **B.** The momentary pitch angle is compared with itself, so the error is zero
- **C.** The momentary pitch angle is held at zero, so the error is zero

25. The autopilot banks the aircraft into a turn and must not let it descend. Why does the elevator have to be deflected as well?

- **A.** Total lift must rise so its vertical part still balances the weight
- **B.** Total lift must fall so its horizontal part still curves the path
- **C.** Total drag must rise so the airspeed stays at the reference value

26. The localizer mode has been selected on its own on the autoflight control panel as an independent lateral mode — which phase is left out?

- **A.** The Capture Phase
- **B.** The Approach-Track Phase
- **C.** The Localizer-Track Phase

27. An aircraft is at 300 ft AGL with the Approach/Land mode engaged and the crew presses the Approach button again — what happens?

- **A.** The mode reverts to the localizer submode
- **B.** The mode disengages immediately
- **C.** The mode stays in operation

28. The decrab is being flown on an automatic landing — which reference is the heading aligned with?

- **A.** The present track
- **B.** The localizer deviation
- **C.** The preselected course

29. The yaw bar in the ADI deflects during the take off run. Which control inputs does the pilot make?

- **A.** Aileron or roll spoiler inputs
- **B.** Rudder or nose wheel steering inputs
- **C.** Elevator or stabiliser trim inputs

30. The aircraft climbs through a radio altitude of 30 ft after lift off. What changes in the flight director?

- **A.** The roll channel is deactivated and the yaw channel activated
- **B.** The yaw channel is deactivated and the roll channel activated
- **C.** The pitch channel is deactivated and the roll channel activated

31. An approach is planned to a decision height of 150 ft with a reported visual range of 400 m — which category applies?

- **A.** CAT II
- **B.** CAT III a
- **C.** CAT I

32. A single autopilot is engaged and it detects its own failure and disengages. Which behaviour does the system show?

- **A.** fail operational behaviour
- **B.** fail passive behaviour
- **C.** fail safe behaviour

33. A single failure occurs below the alert height and degrades the failure mode from fail operational to fail passive. What happens to the displayed capability?

- **A.** it is no longer changed
- **B.** it drops by one category
- **C.** it is replaced by a red warning

ROTARY WING AERODYNAMICS

34. Two reference dimensions of a rotor blade are quoted, one measured along the blade centreline and one across the blade section. Which pair is correct?

- **A.** Span along the centreline, chord across the section
- **B.** Span along the centreline, root across the section
- **C.** Chord along the centreline, span across the section

35. A helicopter rotor turns counterclockwise. The tip-path plane must tilt backward for rearward flight. Following the precession rule, where must the downward force be applied?

- **A.** On the left side
- **B.** Right at the rear
- **C.** On the right side

36. Gyroscopic precession is

- **A.** The change of the rotational axis as a result of an external force
- **B.** The change of the rotor RPM as a result of an external force
- **C.** The turning of the fuselage opposite to the rotor

37. An engine fails and the helicopter enters autorotation. What control does the pilot have?

- **A.** No control, and the helicopter lands where the descent takes it
- **B.** Full control, including the choice of the touchdown point
- **C.** Pedal control only, with no choice of touchdown point

38. During hover, tail rotor thrust makes the helicopter drift to the right. How must the main rotor disk be tilted to prevent this?

- **A.** Slightly forward
- **B.** Slightly to the right
- **C.** Slightly to the left

EQUIPMENT AND FURNISHINGS

39. The same power source that feeds the public address system also feeds other loads that stay powered when every other source is inoperative. How must those other loads be treated?

- **A.** Those other loads are taken into account in the powering duration
- **B.** Those other loads are transferred to the crew interphone supply
- **C.** Those other loads are shed for the whole powering duration

40. Video, announcements and recorded audio all have to reach a seat. Which subsystem controls and distributes them to it?

- **A.** The Passenger Services System
- **B.** The Passenger Entertainment System
- **C.** The Cabin Interphone System

Answer key

Answer, explanation and the manual page each question comes from

1. **B** The middle top position of the Basic T carries the attitude indicator, and the ADI is the attitude indicator that also shows the flight director indication. (L3, PDF p. 361)
2. **A** One of the listed items of RVSM equipment is an autopilot able to maintain the altitude, and that is the function lost. (L3, PDF p. 377)
3. **A** Tip travel is proportional to the applied pressure. (L2, PDF p. 389)
4. **C** A self-sealing coupling closes the line itself the moment it is parted. (L1, PDF p. 423)
5. **B** With an unpressurized cabin the leak lowers the pressure in the static lines, so the altimeter reads slightly higher, while cabin pressure entering the lines of a pressurized aircraft raises that pressure, so it reads lower. (L3, PDF p. 427)
6. **A** At constant altitude neither pressure changes, so the reading is frozen whatever the thrust. (L3, PDF p. 439)
7. **C** The temperature display is used among other things for determining the TAS. (L1, PDF p. 489)
8. **B** Full ram rise gives the highest value at the TAT probe, RAT loses part of it, and SAT has none. (L3, PDF p. 491)
9. **C** The mixture of analogue and digital outputs is what makes the computer hybrid. (L2, PDF p. 513)
10. **A** The module feeds the ADIRU, which is the computer that produces the air data outputs for the displays and other systems. (L3, PDF p. 539)
11. **C** The tube senses the angle as a pressure difference between its two rows of holes. (L2, PDF p. 551)
12. **A** Recording, storing and wireless transmission are the three stated purposes. (L1, PDF p. 613)
13. **C** Components the crew cannot watch are exactly the ones whose overspeed the probe would otherwise miss. (L3, PDF p. 615)
14. **B** Electrical gyros in commercial aircraft carry automatic drift compensation. (L2, PDF p. 715)
15. **C** The remaining deviation goes on the deviation card that stays on the aircraft. (L2, PDF p. 741)
16. **C** Green and blue are the two colours for normal operation, green for a backup system and blue for a temporarily used system. (L3, PDF p. 860)
17. **C** A level 2 caution is given with a single chime. (L2, PDF p. 881)
18. **C** A recorder fault is a level 1 with no aural alarm or warning light, while a stall warning is a level 3 with an aural alarm and a flashing master warning light. (L3, PDF p. 883)
19. **A** Exceeding an envelope limit produces aural messages together with visual annunciations and displays. (L3, PDF p. 903)
20. **C** Rotor rpm is the small R pointer on the inner scale. (L2, PDF p. 963)
21. **B** Flight condition parameters belong to the rotary degrees of freedom. (L1, PDF p. 76)
22. **A** Basic modes normally hold flight condition parameters, and vertical speed is not one. (L3, PDF p. 94)
23. **B** Disengagement is made conditional on both the pilot's input and the design of the autopilot. (L2, PDF p. 96)
24. **B** Both inputs to the summing point are the same signal, so the error is always zero. (L3, PDF p. 106)
25. **A** Banking tilts the lift, so total lift must be increased to keep the vertical part equal to the weight. (L3, PDF p. 142)
26. **B** Selecting LOC as an independent lateral mode omits the Approach-Track Phase. (L3, PDF p. 190)
27. **C** Below 400 ft the button press no longer deactivates the mode. (L2, PDF p. 200)
28. **C** The align phase flies the heading onto the preselected course. (L2, PDF p. 218)
29. **B** Yaw bar commands are flown with rudder or nose wheel steering. (L2, PDF p. 232)
30. **B** At 30 ft the yaw channel drops out and the roll channel comes in at the same moment. (L3, PDF p. 240)
31. **A** A decision height between 200 ft and 100 ft with at least 300 m visual range is CAT II. (L3, PDF p. 272)
32. **B** A single system able to detect and remove its own failure gives fail passive behaviour in single operation. (L3, PDF p. 288)
33. **A** Below the alert height the capability is frozen even though the failure mode degrades. (L3, PDF p. 304)

34. **A** Span runs root to tip along the centreline, and the chord runs leading edge to trailing edge across the section. (L3, PDF p. 6)

35. **A** The output is wanted at the rear; 90° earlier in counterclockwise rotation is the left side. (L3, PDF p. 24)

36. **A** Precession is a change of the rotational axis caused by an external force. (L1, PDF p. 24)

37. **B** During an autorotative landing the helicopter stays fully controllable and the pilot determines the location for touchdown. (L2, PDF p. 36)

38. **C** The disk is tilted opposite to the drift, so to the left for a right drift. (L3, PDF p. 56)

39. **A** The duration must be achieved while considering the other loads left on the same source. (L3, PDF p. 332)

40. **B** All three are routed to the seat through the Passenger Entertainment System. (L2, PDF p. 336)